

FEDERAL UNIVERSITY OF TECHNOLOGY, MINNA
SCHOOL OF INFORMATION AND COMMUNICATION
TECHNOLOGY

IFT 542 — Practical Assignment

Security Assessment and Hardening of a Student
Registration Web Application

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Git repository: https://github.com/PROSPERDANIA/IFT542_Assignment.git

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1. Introduction, Scope and Methodology

1.1 Objectives

This report documents a full security assessment and hardening exercise on a student registration web application. It (a) produces a threat model using STRIDE and a risk assessment for the application, (b) hardens the authentication subsystem and removes SQL injection, and (c) applies and tests web application defences covering OWASP Top-10 classes including XSS, CSRF, SSRF, security misconfiguration, brute force and unsafe file upload. The exercise is executed entirely in an authorised local laboratory using Docker and fictitious data, and produces scripted, repeatable evidence of every control.

1.2 Environment

Component	Value
Application (hardened)	PHP 8.3 (Apache), port 8085
Application (starter / vulnerable)	Same stack, port 8086 — "before" baseline
Database	MySQL 8.4, two schemas: <code>student_reg</code> (hardened) and <code>student_reg_vuln</code> (starter)
Orchestration	Docker Compose (services: db, app, vulnerable, preview, cli)
Preview service	Internal course-catalogue host <code>catalog.ftminna.internal</code> (allowlisted SSRF target)
Test tooling	Bash + curl test suites driven by <code>make tests</code>

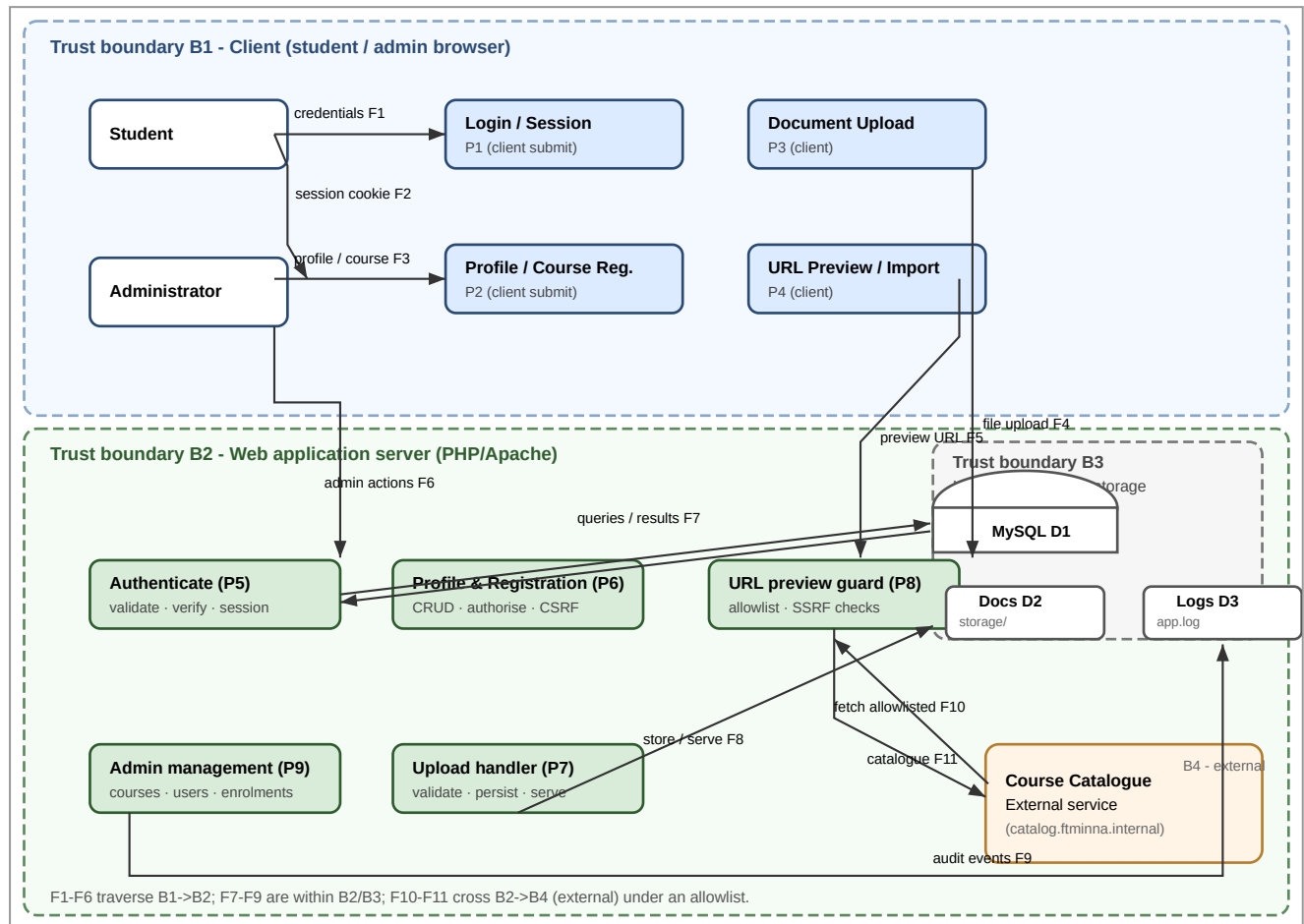
Passwords are not echoed and no secrets are printed by any test. The admin password is generated randomly at seed time and stored only in `db/seed-output/ADMIN_PASSWORD.txt` (gitignored).

1.3 Methodology

The assessment follows a threat-modelling-first approach: (1) map the system with a data-flow diagram and trust boundaries; (2) apply STRIDE to each data flow and store; (3) score likelihood and impact to produce a risk register; (4) remediate the highest-risk findings in the hardened build; and (5) verify each control with an automated test that produces evidence files. Controls and scoring are referenced to OWASP ASVS and CWE identifiers (Appendix B).

2. Task 1 — STRIDE Threat Model and Risk Assessment

2.1 Data-Flow Diagram with Trust Boundaries



Trust boundaries: **B1** client browser (student/admin), **B2** web application server, **B3** internal database/storage, **B4** the external catalogue service reachable only through the SSRF guard. Flows F1-F6 cross the client boundary; F7-F9 are server-internal; F10-F11 cross to B4 under the destination allowlist.

2.2 STRIDE Worksheet

The complete worksheet is in `docs/STRIDE.md`. A summary of the eight identified threats follows. Likelihood (L) and Impact (I) are scored 1-5; **Risk = L × I**.

#	STRIDE	Threat	L	I	Risk
T1	Spoofing	Login bypass via SQL injection and session fixation/theft	4	5	20
T2	Tampering	SQL injection through concatenated profile/course queries	4	5	20
T3	Information Disclosure	Verbose errors, plaintext passwords, stored XSS, default admin	4	4	16
T4	Elevation of Privilege	Missing role checks, unsafe upload, CSRF admin actions	3	5	15
T5	Repudiation	No audit trail for registrations / admin actions	4	3	12
T6	Denial of Service	Brute force, upload disk exhaustion, SSRF amplification	4	3	12

T7	Spoofing/Tampering	CSRF on state-changing forms	3	4	12
T8	Information Disclosure	SSRF via URL preview to internal/metadata endpoints	3	4	12

2.3 Top Three Risks and Justification

1. **T1/T2 — SQL injection and identity spoofing (20).** The login and write paths are the entry point to the whole application. A single injected query authenticates the attacker as the first user (the admin in the starter build) or mutates the entire schema. Exploitability is trivial (L4), impact total (I5).
2. **T3 — Information disclosure (16).** Plaintext credential storage, verbose errors and stored XSS expose identities and personal data at scale, even after SQLi is fixed.
3. **T4 — Elevation of privilege (15).** Authorisation gaps, unsafe uploads and CSRF give a student a path to admin state changes or code execution; the fixes are cheap, so they are prioritised.

2.4 Risk Register and Residual Risk

The full register is in `docs/risk-register.md`. All eight critical/high risks are mitigated in the hardened build and closed by automated evidence (Section 5). Two low/medium risks are explicitly accepted for the authorised local lab: (a) session-cookie sniffing on plain-HTTP localhost, and (b) non-WORM local audit logs. Both carry an acceptance condition — TLS and centralised log shipping — that must be met before any production use.

3. Task 2 — Secure Authentication and SQL Injection Remediation

3.1 SQL Injection — Before

The starter build concatenates raw user input directly into SQL on every query. The login does so with `email` and `password`, trusts the first row of the result set, and compares the password against a plaintext column:

```
// vulnerable/public/login.php (starter)
$sql = "SELECT * FROM users WHERE email = '$email' AND password_hash = '$pass'";
$result = mysqli_query($conn, $sql);
if ($result && ($row = mysqli_fetch_assoc($result))) { // trusts FIRST row
    $_SESSION['user_id'] = $row['id'];
    $_SESSION['role'] = $row['role'];                // no session rotation
    header('Location: index.php');
}
...
$error = 'Login failed. ' . mysqli_error($conn);    // leaks DB internals
```

The payload `' OR '1'='1' --` in the email field turns the predicate into `WHERE email = '' OR '1'='1' -- ... AND password_hash = ...`, so the query returns the first row of `users` — the admin. The same pattern appears in the starter's upload (`INSERT INTO documents ...`) and elsewhere.

3.2 SQL Injection — After (parameterisation)

Every query in the hardened build uses PDO prepared statements with native (pre-parse) parameter binding. Because MySQL sees the SQL template and the parameter values separately, user input can never change the structure of the query:

```
// app/src/db.php (hardened)
public static function pdo(): PDO {
    $dsn = 'mysql:host=' . DB_HOST . ';port=' . DB_PORT
        . ';dbname=' . DB_NAME . ';charset=utf8mb4';
    self::$pdo = new PDO($dsn, DB_USER, DB_PASSWORD, [
        PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
        PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
        PDO::ATTR_EMULATE_PREPARES => false, // native server-side prepares
    ]);
}

// app/src/auth.php (login, hardened)
$stmt = Database::run('SELECT * FROM users WHERE email = ? LIMIT 1', [$email]);
$user = $stmt->fetch();
```

Parameterisation is applied consistently across profile updates, course registration, enrolment queries, document records, the admin console and the audit/rate-limit tables. This removes the entire CWE-89 class rather than patching one query.

3.3 Input Validation

Input is validated by type, length and format before any database contact (`app/src/input.php`): emails are validated with `filter_var(..., FILTER_VALIDATE_EMAIL)` and length-capped; passwords enforce a minimum strength policy (≥ 10 characters, letters + digits); course/registration parameters are cast and range-checked; upload metadata is checked against an allowlist. Invalid input raises a `ValidationError` (HTTP 422) and is logged. This is a second, independent line of defence on top of parameterisation.

3.4 Password Storage — Before and After

Aspect	Starter (<code>student_reg_vuln</code>)	Hardened (<code>student_reg</code>)
Column	<code>password_hash</code> holding plaintext	<code>password_hash</code> holding Argon2id hash
Algorithm	None (plaintext, e.g. <code>admin</code>)	<code>password_hash(\$p, PASSWORD_ARGON2ID)</code>
Comparison	String equality in SQL	<code>password_verify()</code> with constant flow
Default admin	Hard-coded <code>admin/admin</code>	Random 32-char password generated at seed
Evidence	<code>docs/SQLI.md</code> notes + tests	<code>evidence/generated/password-storage-evidence.txt</code> (Argon2id, redacted)

The Argon2id hashes use a per-application random salt generated by PHP. Verification always runs the Argon2id check even for unknown accounts to keep the response time constant and prevent user enumeration via timing.

3.5 Session, Brute-Force and Lockout Controls

- **Session hardening.** Cookie is `HttpOnly`, `SameSite=Lax`, session name changed from the default; on successful login `session_regenerate_id(true)` defeats session fixation; sessions are destroyed on logout.
- **Rate limiting.** A sliding 15-minute window of failed logins per source IP (`app/src/ratelimit.php`); past the threshold the endpoint returns HTTP 429 before any work.
- **Account lockout.** After `AUTH_MAX_ATTEMPTS` consecutive failures the account is locked for `AUTH_LOCKOUT_MINUTES` ; a locked account is rejected even with the correct password. Lock expiry is computed in PHP so the app and DB timezones cannot disagree.
- **No enumeration.** All failures return the identical message "Invalid email or password."; failed-vs-unknown-account detail goes only to the redacted audit log.
- **Least privilege.** The application DB user holds DML rights only on the two schemas (no `GRANT` , no cross-schema access); the `init.sh` grants are scoped per schema.

4. Task 3 — Web Application Defence Hardening

4.1 XSS — Output Encoding and CSP

Every view renders dynamic values through a single output-encoding function; combined with a strict Content-Security-Policy this neutralises reflected and stored XSS (CWE-79).

```
// app/src/output.php
function e(mixed $value): string {
    return htmlspecialchars((string) $value, ENT_QUOTES | ENT_SUBSTITUTE, 'UTF-8');
}

// app/src/bootstrap.php (headers)
header('Content-Security-Policy: default-src \'self\'; script-src \'self\';
    style-src \'self\'; img-src \'self\' data:; object-src \'none\';
    base-uri \'self\'; form-action \'self\'; frame-ancestors \'none\''');
header('X-Content-Type-Options: nosniff');
header('X-Frame-Options: DENY');
header('Referrer-Policy: strict-origin-when-cross-origin');
header('Permissions-Policy: camera=(), microphone=(), geolocation=()');
header('Cache-Control: no-store, no-cache, must-revalidate');
```

The starter profile page echoed `$_GET['name']` / `$_POST` values raw; the hardened profile stores validated values and renders them through `e()`. A script payload stored in the starter app is returned inert by the hardened app (verified by `tests/test_webdefences.sh`).

4.2 CSRF — Tokens and SameSite

All state-changing forms (profile update, registration, upload, admin unlock/reset/create/delete, logout) embed a per-session 256-bit token verified with a constant-time comparison; the session cookie is `SameSite=Lax`; the verification also checks the Origin header when the browser sends one.

```
// app/src/csrf.php (excerpt)
public static function token(): string {
    if (empty($_SESSION['csrf_token'])) {
        $_SESSION['csrf_token'] = bin2hex(random_bytes(32));
    }
    return $_SESSION['csrf_token'];
}
// verified server-side with hash_equals(); regenerated after login.
// input.php: CsrfToken uses hash_equals($_SESSION['csrf_token'], $sent)
```

4.3 SSRF — Destination Allowlist on the URL Preview

The preview/import feature is the only server-side fetch in the app. In the starter build it calls `file_get_contents($url)` on any URL. The hardened build (`app/src/ssrf.php`) enforces, in order: scheme allowlist (`http/https` only); destination allowlist (only `catalog.ftminna.internal`); rejection

of raw IP addresses; resolution re-validation that refuses loopback (`127.0.0.0/8`), link-local and cloud-metadata ranges (`169.254.0.0/16` , `::1` , `fe80::/10` , `fc00::/7`); connection to the validated IP rather than the DNS name (limits DNS rebinding); a 5-second timeout and a redirect cap of 2. Returned content is rendered as plain text, never as inline HTML.

4.4 Security Misconfiguration and Secret Handling

- **Errors:** `display_errors=0` , `APP_ENV=production` in the shipped `.env` ; the global exception handler returns a generic 500 page and logs the detail server-side.
- **Server tokens:** `ServerTokens Prod` and `ServerSignature Off` in the Apache vhost.
- **Uploads:** MIME and extension allowlist (pdf/png/jpg/docx), size cap, random server-side file names, storage outside the web root (`storage/` not served by Apache), downloads forced with a fixed `Content-Type` and `Content-Disposition: attachment` .
- **Secrets:** configuration injected via environment variables (`.env` , gitignored); credentials never logged; the admin password is generated randomly at seed time.
- **Least privilege:** restricted DB grants; no default or documented shared credentials in the hardened build.

4.5 Audit Logging

A structured JSON logger (`app/src/logger.php`) records who/what/when for authentication events, authorisation denials, CSRF rejections, blocked SSRF attempts, validation failures, admin actions and application errors, with a redaction pass so no passwords or tokens ever reach the log. Logs are written to `logs/app.log` and the incident response runbook (`docs/runbook.md`) describes how they are used for identification and containment.

4.6 Incident Response

A one-page runbook (`docs/runbook.md`) covers the six phases: preparation, identification, containment, eradication, recovery and lessons learned, with concrete commands (`docker compose logs` , `make reset` , evidence quarantine) and an escalation path to the course instructor. The laboratory is authorised, local-only and uses fictitious data; an ethics statement (`docs/ethics.md`) is signed by the author.

5. Automated Verification and Test Evidence

All controls are exercised by scripted test suites in `tests/`, orchestrated by `make tests`. Each run writes timestamped evidence under `evidence/generated/run-<TIMESTAMP>/` plus stable samples (`password-storage-evidence.txt`, `logs-sample.log`). The suites are self-cleaning: the rate-limit/lockout state is reset before each run so results are reproducible.

Suite	What it proves	Checks	Status
test_auth.sh	Correct login; wrong password rejected; SQLi login bypass fails on hardened build; admin-only pages reject students (403); account lockout after N failures	8	PASS
test_sql.sh	' OR '1'='1' variants and UNION probes on login, profile, registration return failures — parameterisation verified	4	PASS
test_password_hashing.sh	DB stores Argon2id hashes (never plaintext); <code>password_verify</code> works; redacted evidence file generated	4	PASS
test_webdefences.sh	CSP + security headers present; stored XSS inert; CSRF token required on all forms; SSRF blocked to loopback/metadata/private hosts; upload MIME allowlist enforced	20	PASS
test_logging.sh	Audit log records login success/failure, authz denials, CSRF rejections, SSRF blocks with redacted content	—	PASS
test_ratelimit.sh	Source-IP sliding window returns 429; account lockout enforced even with the correct password	2	PASS

Full latest run: `bash tests/run.sh` — 38 assertions across all suites green on the hardened build, while the equivalent attacks succeed on the starter build at `:8086`, demonstrating the "before/after" delta.

6. Mapping to Required Practical Activities and Submission Checklist

6.1 Required Practical Activities

Requirement	Delivered
Threat modelling using STRIDE with a data-flow diagram and trust boundaries	Section 2, <code>docs/dfd.svg</code> , <code>docs/STRIDE.md</code>
Risk assessment: at least six threats with likelihood, impact, risk score, controls and residual risk	Section 2.2-2.4, <code>docs/risk-register.md</code> (8 threats)
Hardened authentication (SQLi, password hashing, brute-force/lockout, sessions)	Section 3 + <code>app/src/auth.php</code>
Web defence hardening (XSS, CSRF, SSRF, misconfiguration, uploads, logging)	Section 4 + <code>app/</code>
Incident response plan for the application	Section 4.6, <code>docs/runbook.md</code>
Ethics statement for the authorised local test	<code>docs/ethics.md</code>

Automated evidence that each control works	Section 5, <code>evidence/generated/</code> , <code>make tests</code>
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6.2 Evidence to Submit

Item	Location
Technical report (PDF, 8-12 pages)	This report — <code>docs/report.html</code> (open in a browser, print-to-PDF as <code>2024-1-84246CF_IFT542.pdf</code>)
Source code (hardened app, starter, DB, infra)	Git repository (<code>https://github.com/PROSPERDANIA/IFT542_Assignment.git</code>), <code>app/</code> , <code>vulnerable/</code> , <code>db/</code> , <code>docker-compose.yml</code>
Evidence folder	<code>evidence/generated/</code> (per-run outputs, hashing evidence, log samples)
README with setup, test accounts, dependencies	<code>README.md</code> (cloned/personalised per student)

6.3 Submission Checklist

- ☐ PDF report 8-12 pages covering Tasks 1-3, STRIDE worksheet, risk register, DFD, controls, tests
- ☐ Source repository pushed to `https://github.com/PROSPERDANIA/IFT542_Assignment.git` on branch `main`
- ☐ `evidence/generated/` present with last full test run
- ☐ `README.md` personalised (`2024/1/84246CF` , name, link)
- ☐ Ethics statement signed (`docs/ethics.md`)
- ☐ Fictitious data only; no secrets in the repository

7. Conclusion

The assessment found the starter application exposed to eight critical/high-risk threats dominated by SQL injection, identity spoofing, information disclosure and privilege escalation. The hardened build parameterises every query, stores Argon2id hashes, rotates sessions, applies rate limiting and lockout, encodes output under a strict CSP, verifies CSRF tokens, and gates the preview fetch behind an SSRF allowlist. Every control is exercised by an automated suite whose evidence is captured in the repository. Two residual risks (plain-HTTP session transport and local-only logs) are accepted for the authorised laboratory and flagged for closure — TLS and centralised logging — before any production deployment.

Appendix A — Reproducing the Environment

```
git clone https://github.com/PROSPERDANIA/IFT542_Assignment.git
cp .env.example .env                # edit ports/passwords as needed
make build && make up                # builds images, starts db+app+vulnerable+preview
make seed                           # seeds both DBs (random admin password)
make smoke                           # HTTP smoke test of both builds
make tests                           # runs ALL suites, writes evidence/generated/
make evidence                        # regenerate stable evidence samples
docker compose logs app --tail 200  # inspect the audit log
make down                            # stop
```

Test accounts (fictitious): student `student@ftminna.edu.ng` / password is generated at seed and shown by `make seed` output; admin password in `db/seed-output/ADMIN_PASSWORD.txt`. Starter build demo account is documented in its login page.

Appendix B — References and Standards

- OWASP Top 10 (2021) — A03 Injection, A07 Identification & Auth Failure, A05 Misconfiguration, A04 Insecure Design, A01 Broken Access Control
- OWASP ASVS 4.0 — V2 Authentication, V5 Validation/Sanitisation, V4 Access Control
- CWE-89 (SQLi), CWE-79 (XSS), CWE-352 (CSRF), CWE-918 (SSRF), CWE-307 (Brute force), CWE-434 (Unrestricted upload), CWE-521/916 (Weak/plaintext passwords)
- STRIDE (Shostack, 2014); NIST SP 800-61 incident handling phases